ABACUS / IMPLEMENTATION 1.0

User manual

Installation, eight-tab application, command-line workflows and troubleshooting

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Implementation and recorded host checks. MATLAB, GUI, Simulink and ML training have not executed on the delivery host.

User manual

Installation and first use

Extract the archive into a writable folder. In MATLAB R2023b or newer, set Current Folder to Aircraft_Engine_Digital_Twin. Install Database Toolbox. Run setup_project, verify_project and run_all_tests. The test runner writes actual Passed/Failed/Incomplete columns to reports/matlab_test_results.csv. Optional toolbox or desktop limitations may produce incomplete tests, which are not passes. Run run_project('gui') to open the application.

The first setup call preserves the provided seeded database and adds only the authoritative project source folders. You can create a separate database with setup_project(fullfile(tempdir,'my_abacus.db')). To use it in subsequent workflows, pass its filename explicitly to DT_Coordinator or EngineAnalyser_v1; calling run_project uses the standard project database.

Eight application tabs

1. Engine Database Browser: select architecture/engine, inspect catalogue and per-parameter provenance. Source confidence appears with the selected engine.

2. Single-Point Analysis: enter geometric altitude in metres, Mach and throttle. Run physics at L4 or request L5. Inspect SI/display metrics, source label, warnings and station total conditions. No output exists before calculation.

3. Performance Envelope: generate altitude/Mach/throttle sweeps and architecture maps. Invalid cells remain flagged. Plot force, power or fuel metrics only where applicable.

4. Digital Twin Comparison: compare engines at a common flight condition and with one comparable metric. Jet TSFC and propeller BSFC are not interchangeable.

5. ML Surrogate Manager: generate gated data, train candidates and inspect actual metrics. Missing validation or unavailable models produces an actionable diagnostic.

6. Mission-Based Engine Selection: choose a preset or edit constraints and normalized weights, apply filtering, then inspect survivors, eliminations and score contributions.

7. Validation Report: execute physics tests and Track A comparison. Unverified secondary ratings remain WARNING. No table is preloaded with invented passes.

8. Settings and Export: change a compatible database, choose single-point SI/Imperial display, export CSV/MAT/PNG/PDF and create an HTML session report. GUI altitude entry and CSV remain SI.

The GUI calls backend functions; it contains no second implementation of the thermodynamic equations. It is provided as .m AppBase source and cannot be opened as a .mlapp in App Designer Design View. Runtime screen layout and callbacks still require desktop MATLAB acceptance.

Command-line example

```
setup_project;
dt = DT_Coordinator;
f = flight_condition(0,0,1);
```

```
r = dt.run(4,f,'L4',true);
disp(format_result_metrics(r,'SI'));
comparison = compare_engines(dt,[4,5],f,'Thrust_N');
mission = EngineSelector(dt,1,[],true);
reportFile = generate_report(r,comparison,mission,table());
delete(dt);
```

Check r.Valid and r.Warnings before interpreting a calculated metric. L1 can evaluate without persistence; L4 stores results and metadata. L5 attempts ML and labels any physics fallback. A separate engine-off convention applies at throttle zero.

Mission presets

Mission_ID: 1 | Mission_Name: Subsonic Commercial Transport | Architectures: Turbofan_HighBPR | Altitude_m: 11000 | Mach: 0.82 | Ceiling_m: 11000

Mission_ID: 2 | Mission_Name: Air Superiority Fighter | Architectures: Turbojet,Turbofan_LowBPR | Altitude_m: 12000 | Mach: 1.2 | Ceiling_m: 12000

Mission_ID: 3 | Mission_Name: Ground Attack / CAS | Architectures: Turbojet,Turbofan_LowBPR | Altitude_m: 5000 | Mach: 0.85 | Ceiling_m: 5000

Mission_ID: 4 | Mission_Name: Maritime Patrol / ISR | Architectures: Turboprop | Altitude_m: 3000 | Mach: 0.45 | Ceiling_m: 3000

Mission_ID: 5 | Mission_Name: STOL Transport | Architectures: Turboprop | Altitude_m: 2000 | Mach: 0.35 | Ceiling_m: 2000

Mission_ID: 6 | Mission_Name: High-Altitude Reconnaissance | Architectures: Turbojet,Turbofan_LowBPR | Altitude_m: 20000 | Mach: 0.8 | Ceiling_m: 20000

Mission_ID: 7 | Mission_Name: Academic Trainer | Architectures: Turboprop | Altitude_m: 1500 | Mach: 0.25 | Ceiling_m: 1500

The six PDF presets plus Trainer are academic screening examples. First apply architecture, flight-envelope, thrust/power, fuel-specific-consumption, weight and ceiling constraints. Only survivors are normalized and scored. An engine violating a hard condition cannot win. If no engine survives, the application says so; if exactly one survives, it reports feasibility without pretending a relative ranking. Identical criterion values score 0.5 and cannot distinguish candidates. Explanations show five weighted contributions. No aircraft drag, range or full mission fuel integration is included.

Optional piston workflow

```
setup_project;
db_seed_piston_extension;
app = run_project('gui');
```

The opt-in seed adds two Assumed/Low catalogue records. This is an extension to the PDF's 11-engine core. Use shaft power, BSFC, fuel flow and equivalent propulsive thrust for this architecture.

Simulink workflow

```
setup_project;
models = build_simulink_models;
out = run_simulink_model('ABACUS_Turbojet',[1;0;0;1]);
```

Input order is engine ID, geometric altitude m, Mach, throttle. The builder makes three connected .slx models, each with a constant operating point, Level-2 MATLAB S-function and workspace sink. The S-function calls the shared physics backend. Delivered .mdl files can be inspected before building. Stop time zero represents one equilibrium evaluation; running a longer Simulink time does not add engine dynamics. Close an existing model before rebuilding; the builder refuses to overwrite an open model's unsaved edits.

Reporting and output locations

run_project('demo') creates a session report from real calculations once MATLAB runs. plots holds exported figures, tables contains CSV grids/analysis, results contains saved sessions and ml_models contains actual trained bundles. reports holds verification evidence and reports. The supplied technical report documents the implementation and includes source listings; it contains no claimed engine validation or experimental findings.

Troubleshooting

If sqlite is unavailable, install/enable Database Toolbox. A release error requires R2023b or later. A schema error requires backing up the old database and deliberately migrating or choosing a new filename; do not delete a user's database blindly. A provenance error means a numerical parameter and its provenance are missing or inconsistent. A validation-required error is expected for ML with the default unverified references. A missing model warning during L5 results in physics fallback. Invalid operating points must be corrected or left invalid, not clipped into a different condition. If a Simulink wrapper fails, inspect the original backend diagnostic and run its corresponding MATLAB case first.